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Subject Code:BCE402

Roll No:

BTECH

(SEM. IV) THEORY EXAMINATION 2025-26

INTRODUCTION TO SOLID MECHANICS

TIME: 3 HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2x7 = 14

QNo.	Question	CO	Level
(a)	Define stress and strain with units.	1	K1
(b)	State Hooke's Law and its limitations.	1	K2
(c)	Define Poisson's ratio and volumetric strain.	1	K2
(d)	What is shear force and bending moment?	2	K1
(e)	Define neutral axis and section modulus.	3	K2
(f)	What are principal stresses?	4	K1
(g)	State torsion equation for circular shafts.	5	K1

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

(a)	A steel rod of diameter 25mm is subjected to a tensile load of 80kN. Calculate stress, strain, and elongation if length = 2m and $E = 200\text{GPa}$.	1	K3
(b)	Draw SFD and BMD for a simply supported beam of span 6m carrying a central load of 30kN.	2	K3
(c)	A rectangular beam ($b = 150\text{mm}$, $d = 300\text{mm}$) is subjected to bending moment of 15kN-m. Find maximum bending stress.	3	K3
(d)	At a point, $\sigma_x = 100\text{MPa}$, $\sigma_y = 50\text{MPa}$, $\tau_{xy} = 30\text{MPa}$. Find principal stresses.	4	K4
(e)	A shaft transmits 25kW at 200rpm. Calculate torque transmitted.	5	K3

SECTION C

3. Attempt any one part of the following:

7 x 1 = 7

(a)	A composite bar consists of steel ($E_s = 200\text{GPa}$, $A_s = 400\text{mm}^2$) and aluminum ($E_a = 70\text{GPa}$, $A_a = 600\text{mm}^2$). If axial load = 60kN, find stresses in each material.	1	K4
(b)	A bar tapers uniformly from 40mm to 20mm over length 1m. Find elongation under 40kN load.	1	K5

4. Attempt any one part of the following:

7 x 1 = 7

(a)	A cantilever beam of length 4 m carries UDL of 10kN/m. Draw SFD and BMD and find maximum moment.	2	K3
(b)	Using moment area method, determine slope at free end of a cantilever beam carrying point load at free end.	2	K4

5. Attempt any one part of the following:

7 x 1 = 7

(a)	Derive bending equation and calculate bending stress for a circular section of diameter 200mm subjected to bending moment of 20kN-m.	3	K4
(b)	Derive shear stress formula and find maximum shear stress in a rectangular beam ($b = 120\text{mm}$, $d = 240\text{mm}$) under shear force of 40kN.	3	K4

6. Attempt any one part of the following:

7 x 1 = 7

(a)	Determine principal stresses and maximum shear stress for $\sigma_x = 120\text{MPa}$, $\sigma_y = 60\text{MPa}$, $\tau_{xy} = 40\text{MPa}$ using analytical method.	4	K5
(b)	A column 3 m long, pinned at both ends, has cross-section 120mm x 120mm. Find Euler's critical load.	4	K4

7. Attempt any one part of the following:

7 x 1 = 7

(a)	A solid shaft transmits 50kW at 150rpm. If allowable shear stress is 60 MPa, find the required diameter of solid shaft.	5	K4
(b)	A thin cylinder of diameter 1m and thickness 10mm is subjected to internal pressure of 2MPa. Find hoop stress and longitudinal stress.	5	K4